LAB - Alloy for Logs

Docker Containers:

```
docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS         PORTS                                          NAMES
574d4bacd736   grafana/grafana-oss    "/run.sh"                2 minutes ago  Up 2 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    grafana
3f5bd71c1af3   grafana/alloy:latest   "/bin/alloy run --se…"   4 minutes ago  Up 4 minutes   0.0.0.0:12345->12345/tcp, [::]:12345->12345/tcp alloy-agent
f119e4ff10db   grafana/loki:latest    "/usr/bin/loki -conf…"   5 minutes ago  Up 5 minutes   0.0.0.0:3100->3100/tcp, [::]:3100->3100/tcp    loki
122f7dce8655   pokfinner/loggerino    "docker-entrypoint.s…"   6 minutes ago  Up 6 minutes   0.0.0.0:3005->3005/tcp, [::]:3005vijayagagla@DESKTOP-FP4OP26vija
```

Verification:

```
vijayagagla@DESKTOP-FP4OP26:~/logs-for-alloy-lab/loggerino$ curl http://localhost:3005/status
{"ok":true,"uptime":222.750796673}vijayagagla@DESKTOP-FP4OP26:~/logs-for-alloy-lab/loggerino$ curl http://localhost:3005/erroc
rl http://localhost:3005/error
{"error":{"message":"This endpoint always bombs","status":500}}vijayagagla@DESKTOP-FP4OP26:~/logs-for-alloy-lab/loggerino$ cucu
rl http://localhost:3005/random
{"value":0.8234489753112082}vijayagagla@DESKTOP-FP4OP26:~/lo# Using your Browsergerino$ # Using your Browser
```

```
{"value":0.8234489753112082}vijayagagla@DESKTOP-FP4OP26:~/lo# Using your Browsergerino$ # Using your Browser
http://localhost:3100/loki/api/v1/query_range?query={app=%22loggerino%22}

# Using Curl
curl -g "http://localhost:3100/loki/api/v1/query_range?query={app=\"loggerino\"}"
-bash: http://localhost:3100/loki/api/v1/query_range?query={app=%22loggerino%22}: No such file or directory
{"status":"success","data":{"resultType":"streams","result":[{"stream":{"app":"loggerino","detected_level":"unknown","env":"dev","filename":"/app/main.log","service_name":"loggerino"},"values":[[
"1772734733420816391","[2026-03-05T18:18:53.327Z] || LEVEL=SERVER_ERROR || STATUS=500 || METHOD=GET || PATH=\"/error\" || MESSAGE=\"This endpoint always bombs\" || STACK=\"Error: This endpoint al
ways bombs | at file:///usr/src/app/app.js:128:44 | at Layer.handleRequest (/usr/src/app/node_modules/router/lib/layer.js:152:17) | at next (/usr/src/app/node_modules/router/lib/route.js:157:13)
| at Route.dispatch (/usr/src/app/node_modules/router/lib/route.js:117:3) | at handle (/usr/src/app/node_modules/router/index.js:435:11) | at Layer.handleRequest (/usr/src/app/node_modules/router
/lib/layer.js:152:17) | at /usr/src/app/node_modules/router/index.js:295:15 | at processParams (/usr/src/app/node_modules/router/index.js:582:12) | at next (/usr/src/app/node_modules/router/index
.js:291:5) | at jsonParser (/usr/src/app/node_modules/body-parser/lib/types/json.js:100:7)\""]]},{"stream":{"app":"loggerino","detected_level":"info","env":"dev","filename":"/app/main.log","servi
ce_name":"loggerino"},"values":[["1772734749379617659","[2026-03-05T18:19:09.372Z] || LEVEL=INFO || STATUS=200 || METHOD=GET || PATH=\"/random\" || RES_TIME_MS=0.525 || LENGTH=28 || REMOTE_IP=::f
fff:172.31.0.1 || USER_AGENT=\"curl/8.5.0\""],["1772734733420840670","[2026-03-05T18:18:53.331Z] || LEVEL=INFO || STATUS=500 || METHOD=GET || PATH=\"/error\" || RES_TIME_MS=4.297 || LENGTH=63 ||
REMOTE_IP=::ffff:172.31.0.1 || USER_AGENT=\"curl/8.5.0\""],["1772734726321420557","[2026-03-05T18:18:46.297Z] || LEVEL=INFO || STATUS=200 || METHOD=GET || PATH=\"/status\" || RES_TIME_MS=14.600 |
| LENGTH=34 || REMOTE_IP=::ffff:172.31.0.1 || USER_AGENT=\"curl/8.5.0\""]]}],"stats":{"summary":{"bytesProcessedPerSecond":58245,"linesProcessedPerSecond":165,"totalBytesProcessed":1409,"totalLin
esProcessed":4,"execTime":0.024191,"queueTime":0.000675,"subqueries":0,"totalEntriesReturned":4,"splits":2,"shards":2,"totalPostFilterLines":4,"totalStructuredMetadataBytesProcessed":32},"querier
":{"store":{"totalChunksRef":0,"totalChunksDownloaded":0,"chunksDownloadTime":0,"queryReferencedStructuredMetadata":false,"queryUsedV2Engine":false,"chunk":{"headChunkBytes":0,"headChunkLines":0,
"decompressedBytes":0,"decompressedLines":0,"compressedBytes":0,"totalDuplicates":0,"postFilterLines":0,"headChunkStructuredMetadataBytes":0,"decompressedStructuredMetadataBytes":0},"chunkRefsFet
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tes":0,"postPredicateRows":0,"postPredicateDecompressedBytes":0,"postPredicateStructuredMetadataBytes":0,"postFilterRows":0,"pagesScanned":0,"pagesDownloaded":0,"pagesDownloadedBytes":0,"pageBatc
hes":0,"totalRowsAvailable":0,"totalPageDownloadTime":0}}},"ingester":{"totalReached":2,"totalChunksMatched":1,"totalBatches":3,"totalLinesSent":4,"store":{"totalChunksRef":0,"totalChunksDownload
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eDecompressedBytes":0,"postPredicateStructuredMetadataBytes":0,"postFilterRows":0,"pagesScanned":0,"pagesDownloaded":0,"pagesDownloadedBytes":0,"pageBatches":0,"totalRowsAvailable":0,"totalPageDo
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und":0,"entriesRequested":0,"entriesStored":0,"bytesReceived":0,"bytesSent":0,"requests":0,"downloadTime":0,"queryLengthServed":0},"result":{"entriesFound":0,"entriesRequested":0,"entriesStored":
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ntriesStored":0,"bytesReceived":0,"bytesSent":0,"requests":0,"downloadTime":0,"queryLengthServed":0}},"index":{"totalChunks":0,"postFilterChunks":0,"shardsDuration":0,"usedBloomFilters":false}}}}
```
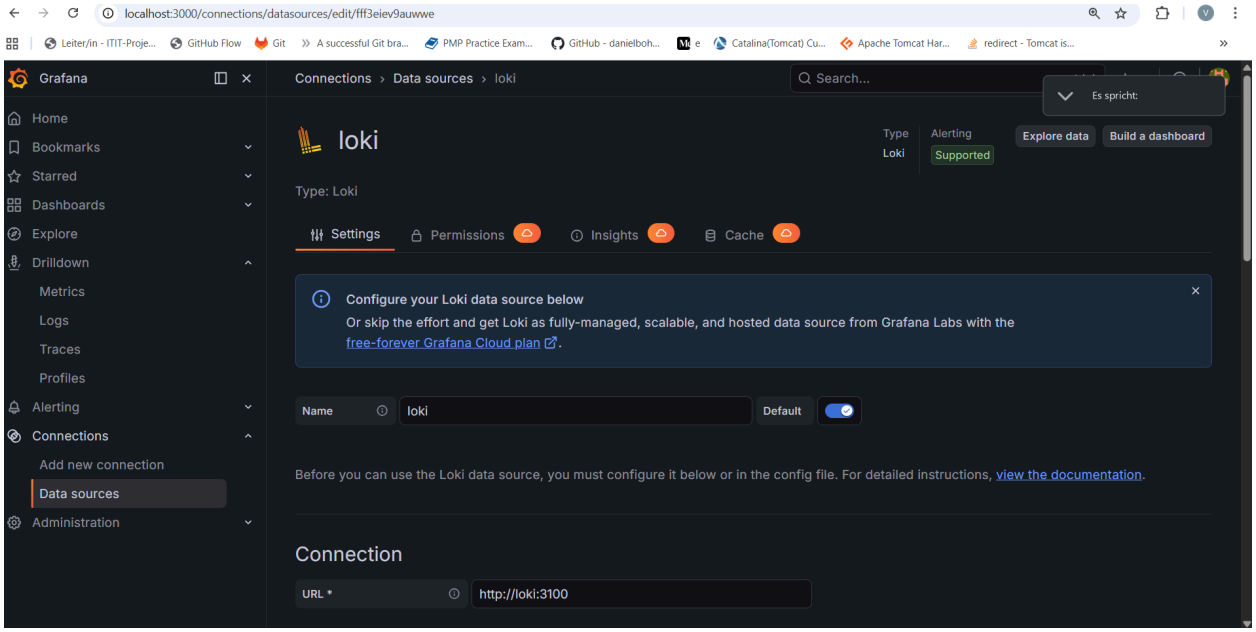
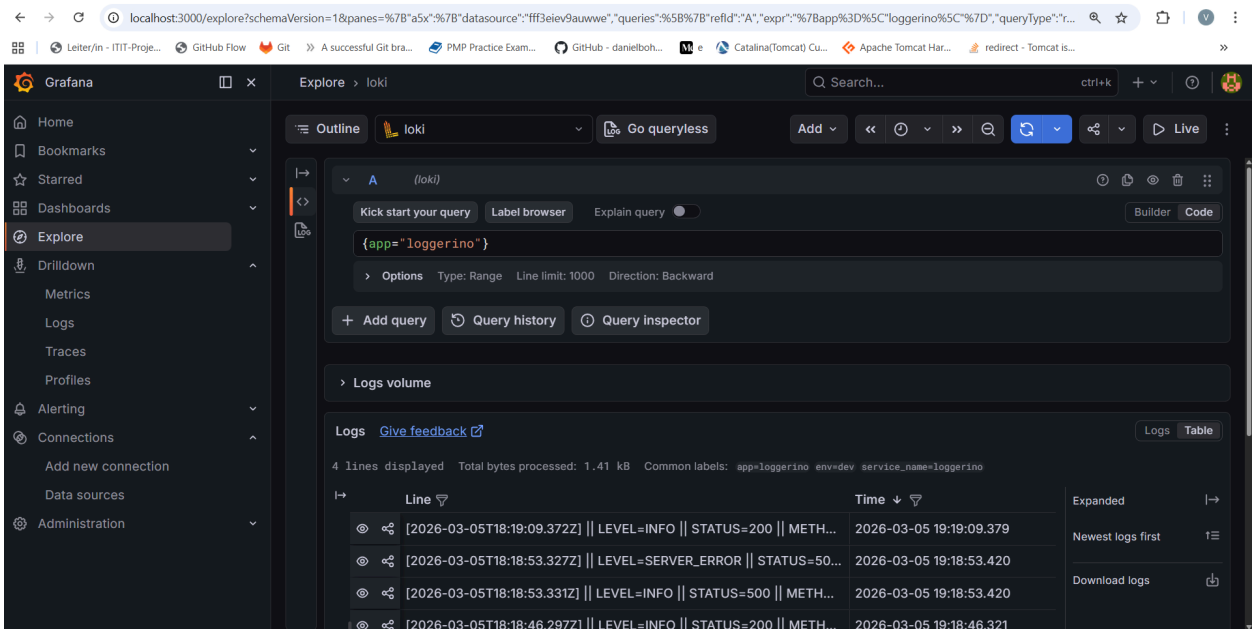
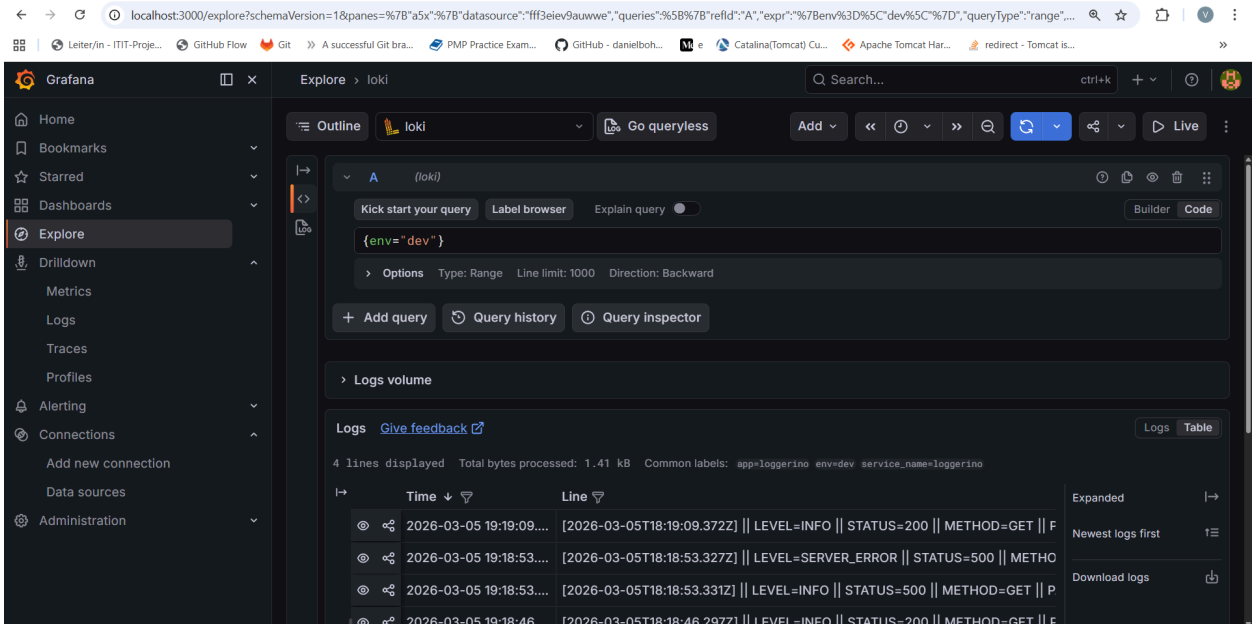

```
vijayagagla@DESKTOP-FP4OP26:~/logs-for-alloy-lab/loggerino$ curl -g "http://localhost:3100/loki/api/v1/query_range?query={app=\"loggerino\"}"
{"status":"success","data":{"resultType":"streams","result":[{"stream":{"app":"loggerino","detected_level":"unknown","env":"dev","filename":"/app/main.log","service_name":"loggerino"},"values":[[
"1772734733420816391","[2026-03-05T18:18:53.327Z] || LEVEL=SERVER_ERROR || STATUS=500 || METHOD=GET || PATH=\"/error\" || MESSAGE=\"This endpoint always bombs\" || STACK=\"Error: This endpoint al
ways bombs | at file:///usr/src/app/app.js:128:44 | at Layer.handleRequest (/usr/src/app/node_modules/router/lib/layer.js:152:17) | at next (/usr/src/app/node_modules/router/lib/route.js:157:13)
| at Route.dispatch (/usr/src/app/node_modules/router/lib/route.js:117:3) | at handle (/usr/src/app/node_modules/router/index.js:435:11) | at Layer.handleRequest (/usr/src/app/node_modules/router
/lib/layer.js:152:17) | at /usr/src/app/node_modules/router/index.js:295:15 | at processParams (/usr/src/app/node_modules/router/index.js:582:12) | at next (/usr/src/app/node_modules/router/index
.js:291:5) | at jsonParser (/usr/src/app/node_modules/body-parser/lib/types/json.js:100:7)\""]]},{"stream":{"app":"loggerino","detected_level":"info","env":"dev","filename":"/app/main.log","servi
ce_name":"loggerino"},"values":[["1772734749379617659","[2026-03-05T18:19:09.372Z] || LEVEL=INFO || STATUS=200 || METHOD=GET || PATH=\"/random\" || RES_TIME_MS=0.525 || LENGTH=28 || REMOTE_IP=::f
fff:172.31.0.1 || USER_AGENT=\"curl/8.5.0\""],["1772734733420840670","[2026-03-05T18:18:53.331Z] || LEVEL=INFO || STATUS=500 || METHOD=GET || PATH=\"/error\" || RES_TIME_MS=4.297 || LENGTH=63 ||
REMOTE_IP=::ffff:172.31.0.1 || USER_AGENT=\"curl/8.5.0\""],["1772734726321420557","[2026-03-05T18:18:46.297Z] || LEVEL=INFO || STATUS=200 || METHOD=GET || PATH=\"/status\" || RES_TIME_MS=14.600 |
```

Loki Datasource :



Loki Sample Queries:




localhost:3000/explore?schemaVersion=1&panes=%7B"a5x":%7B"datasource":"fff3eiev9auwwe","queries":%5B%7B"refId":"A","expr":"%7Benv%3D%5C"dev%5C"%7D","queryType":"range",...
Grafana
Explore › loki
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Connections
Add new connection
Data sources
Administration
Outline
loki
Go queryless
Add
Live
A (loki)
Kick start your query
Label browser
Explain query
Builder
Code
{env="dev"}
Options
Type: Range
Line limit: 1000
Direction: Backward
Add query
Query history
Query inspector
Logs volume
Logs
Give feedback
Logs
Table
4 lines displayed
Total bytes processed: 1.41 kB
Common labels:
app=loggerino
env=dev
service_name=loggerino
Time
Line
2026-03-05 19:19:09....
[2026-03-05T18:19:09.372Z] || LEVEL=INFO || STATUS=200 || METHOD=GET || F
2026-03-05 19:18:53....
[2026-03-05T18:18:53.327Z] || LEVEL=SERVER_ERROR || STATUS=500 || METHO
2026-03-05 19:18:53....
[2026-03-05T18:18:53.331Z] || LEVEL=INFO || STATUS=500 || METHOD=GET || P
Expanded
Newest logs first
Download logs